VAIBHAV GARG

Software Engineer | Data Engineer | Python | PySpark | AWS | Snowflake | SQL | ML

vaibhavgargindia01@gmail.com | +91 74529 83652 | linkedin.com/in/vaibhavgarg003 | Open to PAN India | Available Immediately

PROFESSIONAL SUMMARY

Software and Data Engineer with 3 years of production experience building Python services, scalable ETL/ELT pipelines, RESTful APIs, and cloud-native data architectures on AWS. Delivered consistent, measurable impact across every engagement:

- **30%** reduction in pipeline execution time through PySpark re-architecture on AWS Glue
- **5+ TB** of operational data processed daily with zero SLA breaches
- **95%+** elimination of production data errors via Python validation frameworks
- **Zero-incident** cloud migration from on-premise Oracle to AWS + Snowflake

Core stack: Python, PySpark, SQL, AWS (S3, Lambda, Glue, EC2, CloudWatch, IAM), Snowflake, GCP, REST APIs, Git, CI/CD, Scikit-learn, Power BI. Master of AI & ML, University of Adelaide (GPA 6.6/7.0, 2025). Open to Software Engineer, Data Engineer, ML Engineer, and Cloud Engineer roles across India.

TECHNICAL SKILLS

Programming Languages: Python (Expert), SQL (Advanced), PySpark, Scala (exposure), Shell Scripting, Java (basic)

Python Ecosystem: Pandas, NumPy, Scikit-learn, SQLAlchemy, Pydantic, FastAPI/Flask (REST APIs), pytest, Matplotlib, Seaborn, Plotly

Big Data & ETL: Apache Spark, PySpark, Hadoop, Hive, ETL/ELT Pipeline Design, Data Ingestion, Data Transformation, Data Quality, Data Lineage

AWS Cloud Services: S3, Lambda, Glue, EC2, Redshift, API Gateway, IAM, CloudWatch, DMS, EMR, SNS, SQS, DynamoDB, Athena

Cloud & Warehousing: Snowflake, GCP (BigQuery exposure), Azure (exposure), Multi-cloud Architecture, Data Lakes, Dimensional Modelling

ML & Data Science: Scikit-learn, Regression, Classification, Time-Series, Decision Trees, Clustering, Neural Networks, Model Deployment

DevOps & CI/CD: Git, GitHub Actions, AWS CodePipeline, Branch Management, Build Pipelines, Docker (exposure), Terraform (exposure)

Databases: PostgreSQL, MySQL, Oracle, Snowflake, DynamoDB — schema design, migrations, query optimisation, indexing

Monitoring & Observability: CloudWatch (metrics, alarms, log groups), Power BI, Structured Logging, SLA Tracking, Root-Cause Analysis

Methodology: Agile, Scrum, SDLC, Code Reviews, Unit & Integration Testing (pytest), Secure Coding, Secrets Management

PROFESSIONAL EXPERIENCE

Software Engineer — Data & Analytics | Impetus Technologies *Aug 2021 – Dec 2023*

Python | PySpark | Apache Spark | AWS (S3, Glue, Lambda, EC2, CloudWatch, IAM) | Snowflake | GCP | REST APIs | SQL | Git | Agile

- Designed and built distributed PySpark ETL pipelines on AWS Glue processing 5+ TB of daily operational data — re-architected partition strategy, eliminated shuffle-heavy joins, and parallelised sequential jobs, achieving **30% faster** execution with zero SLA breaches post-deployment.
- Executed **zero-incident** migration of on-premise Oracle databases to AWS (S3, Glue, Lambda) and Snowflake — designed target dimensional schemas, built Python validation and reconciliation scripts, and onboarded downstream BI teams onto automated dashboards within 8 weeks.
- Built Python data-quality validation frameworks enforcing schema integrity, null/duplicate detection, and referential consistency at every AWS Glue ingestion stage — reduced production data errors by **95%+**, with zero downstream rework requests after implementation.
- Developed and deployed **RESTful APIs** using Spring Boot with role-based IAM-equivalent authentication, input validation, serialisation, and structured error handling — processed thousands of daily requests with zero unauthorised access incidents.

- Monitored production AWS environments using **CloudWatch** — configured custom metrics, alarms, and log groups; reduced pipeline failure detection time from hours to under 15 minutes; maintained runbooks and SOPs in Git.
- Delivered **ML models** (classification, regression, time-series, clustering) using Python, Scikit-learn, Pandas, and NumPy — translated business problems into data science solutions for cross-functional client teams across Retail, Finance, and Operations.
- Operated within Git-based **CI/CD** workflows across Agile sprints with international teams — managed feature branches, participated in code reviews, facilitated daily scrums, and prepared weekly leadership reports.

Console Operator (Responsible Person) | **OTR Group — Viva Energy** *Apr 2024 – Present*
Power BI | Operational analytics | Incident management | SLA compliance | Team coordination

- Monitored real-time store performance metrics via Power BI dashboards — identified anomalies, applied structured resolution steps, and documented outcomes. Transferable directly to CloudWatch-based production monitoring.
- Managed shift-level incidents and service requests — prioritised competing issues under pressure, applied SOPs, and maintained operational compliance. Demonstrates the accountability and structured problem-solving discipline software engineering roles require.

KEY PROJECTS

Pelmorex — Cloud-Native Data Pipeline Migration | AWS (S3, Glue, Lambda, EC2) · PySpark · Snowflake · GCP · Python
North American weather data platform — full migration from fragmented on-premise Oracle systems to multi-cloud cloud-native architecture.

- Built Python orchestration scripts to manage data availability signals across AWS, Snowflake, and GCP — enforced cross-environment schema consistency, triggered Lambda functions downstream, and automated reconciliation reports. Reduced manual ops by 30%, delivered analytics data 2x faster.
- Re-architected PySpark transformation jobs: redesigned time-series partition keys, eliminated wide shuffles, parallelised Glue job chains — 30% execution time reduction, zero SLA misses post-deployment.
- Migrated analytical datasets from Oracle to Snowflake — designed dimensional schemas, built Python-based row count and integrity validation, and delivered signed-off reconciliation reports to business stakeholders.

Risk Analytics System — RESTful Python Service | Python · Spring Boot · REST APIs · SQL · Authentication

- Designed and implemented a REST API layer with role-based authentication, input validation, serialisation, and error handling — powered a loan eligibility ML scoring engine processing thousands of decisions daily. Zero security incidents. Full SDLC ownership from requirements to production.

EDUCATION

Master of Artificial Intelligence & Machine Learning *Jan 2024 – Dec 2025*
University of Adelaide, Australia | GPA: 6.6 / 7.0

- Coursework: Deep learning, neural networks, regression, time-series, ML model deployment, cloud data systems, NLP, computer vision.
- Adelaide Graduate Award (Nov 2025) — university-recognised for leadership, professional excellence, and cross-cultural collaboration.

Bachelor of Computer Science *2017 – 2021*
Rajasthan Technical University, India | 80% — B.E./B.Tech equivalent

CERTIFICATIONS

AWS Fundamentals Specialization — Amazon Web Services *Sep 2020*
Advanced Microsoft Power BI — Edureka *Mar 2026*
Python for Everybody Specialization — University of Michigan *Jul 2020*
AWS Solutions Architect Associate — *Currently self-studying, target Q2 2026*

PROFESSIONAL REFEREES

Arun Nautiyal — Team Lead, Impetus Technologies | arunnautiyal.a@gmail.com
Keerthan Francis — Store Manager, OTR Norwood (Viva Energy) | dominic.francis.k@gmail.com